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### Connection equipment adapter for a wind turbine tower section

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#### Abstract

The following describes a supporting system for supporting a tower section of a wind turbine. The supporting system includes a connection plate including a tower connection portion and a support connection portion, wherein the support connection portion is configured for being coupled to a tower support device for supporting the tower section. The tower connection portion includes a tower connection hole for being coupleable to a tower hole extending through a shell section of the tower section. The supporting system further includes a clamping device being partially insertable in the tower connection hole of the tower connection portion and the tower hole for detachably clamping the connection plate to the tower section.

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## Background/Summary

## CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims priority to EP application Ser. No. 22/162,745.8, having a filing date of Mar. 17, 2022, the entire contents of which are hereby incorporated by reference.

## FIELD OF TECHNOLOGY

(2) The following relates to a supporting system for supporting a tower section of a wind turbine and a tower section supporting arrangement. Furthermore, the following relates to a method of supporting a tower section of a wind turbine.

## BACKGROUND

(3) A tower of a wind turbine is made of the plurality of tower sections. The tower sections are transported independent from each other to the wind turbine site. Conventional tower sections comprise at axial end faces a plurality connection holes. The connection holes extend along an axial direction of the tower sections and the wind turbine tower, respectively. The end faces of the tower sections are equipped for example with L-flanges or T-flanges through which the axial connection holes extend. Respective connecting bolts and screws are inserted in the connection holes of two adjacent tower sections in order to provide a fixation between the two adjacent tower sections.

(4) The axial connection holes may be used to fix the tower sections during transportation by bolts to a Universal Tower Transport Equipment (UTTE). However, modern turbine sections may not comprise connection holes in the axial end faces so that a connection to a UTTE by respective connecting bolts may not be possible. Instead, the tower sections comprise tower holes that are formed in a shell section of the tower section instead of a front face of the tower section. Thus, two tower section comprise overlapping shell sections such that the respective tower holes in the shells overlap as well. Next, a specific wedge flange coupling system is inserted into the respective overlapping tower holes of the tower sections to be connected. The tower holes have large oval-shaped holes, into which a fastener can be inserted. Furthermore, two wedge shaped clamping jaws are inserted into the tower holes, wherein the clamping jaws can be pressed apart from each other by the fastener. Hence, the fastener clamps both tower sections together by means of the two clamping jaws, that can be pressed against sidewalls in the tower holes and thus creates a symmetrical prestressed connection.

## SUMMARY

(5) An aspect relates to a system for connecting wind turbine sections having tower holes in the shell to a tower transport equipment.

(6) A supporting system for supporting a tower section of a wind turbine, a tower section supporting arrangement and a method of supporting a tower section of a wind turbine is described.

(7) According to a first aspect of embodiments of the present invention, a supporting system for supporting a tower section of a wind turbine is presented. The supporting system comprises a connection plate comprising a tower connection portion and a support connection portion, wherein the support connection portion is configured for being coupled to a tower support device for supporting the tower section. The tower connection portion comprises a tower connection hole for being coupleable to a tower hole extending through a shell section of the tower section. The supporting system further comprises a clamping device being partially insertable in the tower connection hole of the tower connection portion and the tower hole for detachably clamping the connection plate to the tower section.

(8) According to a further aspect of embodiments of the present invention, a tower section supporting arrangement is presented. The tower section supporting arrangement comprises a tower section comprising a tower hole extending through a shell section of the tower section. furthermore, the arrangement comprises a tower support device configured for supporting the tower section, and the above-described system. The support connection portion is coupled to the tower support device, wherein the clamping device is partially inserted in the tower connection hole of the tower

connection portion and the tower hole for detachably clamping the connection plate to the tower section.

(9) According to a further aspect of embodiments of the present invention, a method of supporting a tower section of a wind turbine is presented. According to the method a support connection portion of a connection plate is coupled to a tower support device. A tower connection portion of the connection plate is coupled with a tower connection hole to a tower hole extending through a shell section of the tower section. Furthermore, according to the method, a clamping device is partially inserted in the tower connection hole of the tower connection portion and the tower hole for detachably clamping the connection plate to the tower section.

(10) A tower section has a cylindrical shape. The tower section comprises a center axis defining an axial direction. Accordingly, perpendicular to the axial direction, a radial direction extends from the center point to the outer shell of the tower section. The circumferential direction is the direction in perpendicular to the radial direction and the axial direction. The tower section supported by the inventive supporting system comprises respective tower holes that are formed in a shell section of the tower section instead of a front face of the tower section. The shell section may comprise for example a welded circumferential flange welded to a lateral end of the skin of the shell. This means, that the tower holes extend along a radial direction through the shell section, i.e., the skin surface, the surface shell and/or the circumferential flange, of the tower section. Accordingly, for mounting the tower section to a support device, it is not necessary that the tower section comprises axially extending holes in a front face of the tower section.

(11) A support device is a device that supports the tower section during transportation, lifting and storage. Therefore, the support device is configured to stabilize the tower section and/or to provide a coupling e.g., to a transportation device.

(12) For example, the support device is a support flange being coupleable to a tower section handling device, in particular a transport vehicle or a support platform. The support flange may have standardized connections with a transport device and thereby forms a Universal Tower Transport Equipment (UTTE).

(13) Furthermore, the support device is a tower stabilization beam extending between two opposite sections of the shell section) of the tower section. The tower stabilization beam thereby enforces the tower section during transportation, lifting and storage. For example, vibrations and other impacts during transportation may be formed the tower section such that there is a risk that during transportation the tower section is deformed in a rather elliptical shape. Therefore, the tower stabilization beam functions as an Anti Ovalization Equipment (AO).

(14) Furthermore, the support device may also provide a coupling to respective standard lashing elements. The lashing elements are used to fix lashing strips to secure the tower section from roll and pitch motions during sea and land transport.

(15) The connection plate has a platelike structure defining a tower connection portion, to which the tower section can be mounted, and a support connection portion, to which the support device can be mounted. The tower connection portion comprises a tower connection hole that can be matched or overlaid with the tower hole, when the tower connection portion is attached to a surface of the shell section of the tower section. In other words, if the tower connection portion arranged onto the shell section of the tower section, both, the tower hole and the connection with forms common through holes for the clamping device and specifically a clamping fastener (such as a screw or bolt) of the clamping device. In other words, the tower connection hole in the tower connection portion extends along a radial direction of the tower section, when the connection plate is arranged onto the shell section of the tower section.

(16) The support connection portion does also comprise a through hole, through which a further clamping fastener (such as a screw or bolt) can be guided for fixing the support device to the support connection portion.

(17) The connection plate may be formed monolithically of one piece. Alternatively, the tower

connection portion and the support connection portion may be made of separate parts. Hence, the tower connection portion and the support connection portion may be e.g., welded together.

(18) The clamping device comprises at least an element, such as the below described clamping fastener or the adapter element which is guidable at least through the tower hole and the tower connection hole. For example, the clamping device comprises respective plate washers that are arranged at opposed surfaces of the shell section of the tower section, wherein the tower connection portion of the connection plate and the shell section of the tower section is clamped between the plate washers. For example, fixing screw may be guided through the plate washers and may clamp the two plate washers together for providing the clamping fixation.

(19) Specifically, the clamping device is configured for detachably mounting the connection plate to the tower section and the support device. Hence, by the supporting system according to embodiments of the present invention, a tower section of a wind turbine, which comprises radial directed tower holes in the shell section, can be detachably connected by the clamping device to a respective (temporary) support device. Hence, by the supporting system a handling and transportation of heavy tower sections is possible.

(20) According to an exemplary embodiment, the clamping device comprises a plate washer arrangeable at a surface of the shell section of the turbine tower. The clamping device further comprises a clamping fastener (for example a threaded bolt or a screw) for clamping the plate washer to the tower connection portion of the connection plate and for thereby clamping the tower section between the plate washer and the tower connection portion. The clamping fastener may be guided through the tower hole, the tower connection hole and a hole in the plate washer. The hole in the plate washer may comprise a respective thread for providing a screw connection with the clamping fastener. Additionally or alternatively, a counter nut may be screwed at an end of the clamping fastener to provide the clamping connection.

(21) According to further exemplary embodiment, the plate washer comprises a curved contact surface for contacting the tower section, wherein the curved contact surface is in particular adaptable to the curvature of the respective surface of the tower section.

(22) According to further exemplary embodiment, wherein the connection portion of the connection plate comprises a curved contact surface for contacting the tower section, wherein the curved contact surface is in particular adaptable to the curvature of the respective surface of the tower section.

(23) By adapting the contact surface of the plate washer and/or the connection portion of the connection plate, the larger and homogeneous contact surface the shell section of the tower section can be provided.

(24) Additionally or alternatively, an intermediate contacting plate can be interposed between the shell section and the connection portion or the shell section and the plate washer. The connecting plate may comprise for example two opposing surfaces, the one surface may be adapted to the curvature of the shell section of the tower section and the other surface may be adapted to the (flat, non-curved) shape of the connection portion or the plate washer. The contacting plate may be made of a deformable and elastic material, such as bonded foam and/or hard rubber. The elastic material is compressed by the clamping device and adopts the tower section curvature shape. The intermediate contacting plate can be glued to the connection portion.

(25) According to a further exemplary embodiment, the clamping device comprises an adapter element being adapted to the cross section of the tower hole of the tower section, wherein the adapter element is insertable in the tower hole. The adapter element comprises a through hole extending between the plate washer and the connection portion of the connection plate for inserting the clamping fastener. The adapter element may form a hollow or solid body which fills the inner volume of the tower hole. Hence, relative movements of the adapter element inside the tower hole and therefore also relative movements between the parts of the clamping device and the connection plate may be reduced and prevented. Therefore, a more robust coupling may be achieved.

(26) According to further exemplary embodiment, the adapter element comprises at least two clamping elements for being clampable in the tower hole. The clamping elements are configured for being adjustable with respect to each other, such that the adapter element is adaptable to the cross section of the tower hole. For example, the clamping elements may be movable with respect to each other. The clamping elements may comprise respective clamping surfaces which may be pressed against the inner surfaces of the tower hole. For example, if the tower hole comprises a larger diameter, the two clamping elements may be spaced apart in order to press the respective clamping surfaces of the clamping elements to the surface of the through hole.

(27) Between the clamping elements spindle (mandrel) may be arranged, wherein upon rotation of the spindle, the clamping element may move together or apart from each other.

(28) Furthermore, the two clamping elements may be movably (e.g., slidably) coupled to a central carrier body of the adapter element. The surface of the central carrier body, to which 1 of the clamping elements is arranged, may have a wedge like (conical extending) surface. Upon sliding the clamping element along the wedge like surface, the distance to the other clamping element may be increased or decreased.

(29) According to further exemplary embodiment, the adapter element comprises an elliptical cross section, i.e., a cross section corresponding to an elongated tower hole) for being inserted in a respective tower hole having a respective elliptical cross section. specifically, the above-described clamping element of the adapter element may be configured to be movable along a major axis of the elliptical cross section of the tower.

(30) According to further exemplary embodiment, the tower connection portion and the support connection portion of the connection plate are arranged in one common plane. Hence, a supporting hole or through opening of the support device may have the same hole extension as the radial hole extension of the tower hole.

(31) According to further exemplary embodiment, the connection plate has as an L-shape and the tower connection portion and the support connection portion are formed perpendicular to each other. Hence, a supporting hole or through opening of the support connection portion may have a hole extension perpendicular to the radial hole extension of the tower hole. Specifically, conventional transport devices, such as the Universal Tower Transport Equipment (UTTE), are designed for being coupled to tower sections, having axial tower holes. Hence, by the L-shaped connection plate and the respective 90-degree angle between the tower connection portion and the support connection portion, the modern tower sections having radial extending tower holes in the tower shell section can be coupled to the conventional transport devices, without modification of the conventional transport devices.

(32) According to further exemplary embodiment, the support device comprises the through opening, wherein the clamping device comprises a further plate washer arrangeable at a surface of the support device. The clamping device further comprises a further clamping fastener (for example a threaded bolt or a screw) for clamping the further plate washer to the support connection portion of the connection plate and for thereby clamping the support device between the further plate washer and the support connection portion. The further clamping fastener may be guided through the support device coupling opening, a supporting hole or through opening in the support connection portion and a hole in the further plate washer. The hole in the further plate washer may comprise a respective thread for providing a screw connection with the clamping fastener. Additionally or alternatively, a counter nut may be screwed at an end of the further clamping fastener to provide the clamping connection.

(33) According to further exemplary embodiment, the further plate washer comprises an elongated hole through which the further clamping fastener is guidable.

(34) Embodiments of this invention serve as an adapter between a variety of different tower equipment and the complex wedge tower sections having along its circumferential direction a plurality of elongated through holes in the shell section of the tower section. Those wedge tower

sections are coupled together by a Wedge Connection Equipment Adapter which provides a robust clamping connection between two tower sections. The supporting system of embodiments of the present invention uses the elongated tower holes in the shell section to attach the wedge tower sections to the current support devices such as the support flange (Universal Tower Transport Equipment frame), the Anti Ovalization equipment, Sea-fastening lashing points as well as the top Tower Brackets used for tower section lifting and tower section upending. By using the same support system across different support device equipment and thereby enabling the reuse conventional support devices, the complexity and costs are reduced.

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## Description

### BRIEF DESCRIPTION

- (1) Some of the embodiments will be described in detail, with reference to the following figures, wherein like designations denote like members, wherein:
- (2) FIG. 1 shows a perspective view of a tower section being clamped to support devices by the supporting system according to an exemplary embodiment of the present invention;
- (3) FIG. 2 shows a detailed view of a tower section being clamped to an interface plate of a support flange **142** according to an exemplary embodiment of the present invention;
- (4) FIG. 3 shows a detailed view of a tower section being clamped to an interface plate of a support flange according to an exemplary embodiment of the present invention;
- (5) FIG. 4 shows a schematic view of an L-shaped connection plate according to an exemplary embodiment of the present invention;
- (6) FIG. 5 shows a schematic view of an adapter element according to an exemplary embodiment of the present invention;
- (7) FIG. 6 shows a schematic view of a supporting system according to an exemplary embodiment of the present invention;
- (8) FIG. 7 shows a detailed view of a further plate washer having an elongated hole according to an exemplary embodiment of the present invention;
- (9) FIG. 8 shows a schematic view of a supporting system which couples a tower stabilization beam to the tower section according to an exemplary embodiment of the present invention; and
- (10) FIG. 9 shows a schematic view of a supporting system which couples a lashing to the tower section according to an exemplary embodiment of the present invention.

### DETAILED DESCRIPTION

- (11) The illustrations in the drawings are schematically. It is noted that in different figures, similar or identical elements are provided with the same reference signs.
- (12) FIG. 1 shows tower section supporting arrangement with a tower section **150** being clamped to support devices **140** by the supporting system according to an exemplary embodiment of the present invention.
- (13) The supporting system supports the tower section **150** of a wind turbine. The supporting system comprises a connection plate **100** comprising a tower connection portion **101** and a support connection portion **102**, wherein the support connection portion **102** is configured for being coupled to the tower support device **140** for supporting the tower section **150**. The tower connection portion **101** comprises a tower connection hole for being coupleable to a tower hole **151** extending through a shell section **152** of the tower section **150**. The supporting system further comprises a clamping device **110** being partially insertable in the tower connection hole of the tower connection portion **101** and the tower hole **151** for detachably clamping the connection plate **100** to the tower section **150**.
- (14) The tower section **150** has a cylindrical shape. The tower section **150** comprises a center axis defining an axial direction a. Accordingly, perpendicular to the axial direction a, a radial direction r

extends from the center point to the outer shell section of the tower section. The circumferential direction c is the direction in perpendicular to the radial direction r and the axial direction a. The tower section supported by the inventive supporting system comprises respective tower holes **151** that are formed in the shell section **152** of the tower section **150** instead of a front face of the tower section. The shell section **152** comprises a welded circumferential flange **213** welded to a lateral end of the skin of the shell section **152**. Hence, the tower holes **151** extend along the radial direction r through the shell section **152**, i.e., the skin surface or surface shell, of the tower section **150**.

(15) A tower support device **140** is a device that supports the tower section **150** during transportation. Therefore, the tower support device **140** is configured to stabilize the tower section **150** and/or to provide a coupling to a transportation device.

(16) As shown in FIG. 1, the support device is a support flange **142** being coupleable to a tower section handling device, in particular a transport vehicle or a support platform. The support flange **142** may have standardized connections with a transport device and thereby forms a Universal Tower Transport Equipment (UTTE). Furthermore, an interface plate **144** of the support flange **142** may provide a coupling area for the tower section **150** so that a plurality of connection plates **100** and clamping devices **110** of the supporting system can be used between the interface plate **144** and the tower section **150** to provide a stable support.

(17) Furthermore, the support device **140** is a tower stabilization beam **141** extending between two opposite sections of the shell section **152** of the tower section **150**. The tower stabilization beam **141** thereby enforces the tower section during transportation. The tower stabilization beam **141** functions as an Anti Ovalization Equipment (AO).

(18) The inventive supporting system enables the tower stabilization beam **141** to be mounted to the connection plate **100**, and thereby ensuring that no excessive ovalization happens on these otherwise very slender and ovalization prone tower sections **150** occur. For mounting the tower stabilization beam **141**, the tower section **150** is lifted in the top to de-ovalize it, next, the adapter bolts (further clamping fasteners **211**) are then installed in the identified support portion holes (this can be done by hand by a technician). Then the tower stabilization beam **141** is lifted into place horizontally, using a forklift or similar, the connection plates **100** and the tower stabilization beam **141** are then bolted together and the tower section **150** is release from lifting.

(19) Furthermore, the support device **140** may also provide a coupling to respective standard lashing elements **143**. The lashing elements **143** are used to fix lashing strips to secure the tower section **150** e.g., from roll and pitch motions during sea and land transport. For mounting, a technician will be lifted to the top of the tower section **150** to insert the clamping fastener **210** together with the connection plate **100**. To the connection plate **100**, the lashing elements **143** are already attached into the designated holes in the support connection portion **102**. The lashing strips are then attached to their brackets and tightened up against anchors in the deck of the ship or in points of the vehicle.

(20) Summarizing, a plurality of connection plates **100** and clamping devices **110** of the supporting system can be used to fix one of the support devices **140** and the tower section **150** to provide a stable support. For example, the tower connection portion **101** of the connection plates **100** can be arranged onto an inner surface of the shell section **152** (see for example the connection of the support flange **142** and the lashings **143**) or onto an outer surface of the tower section **150** (see for example the connection of the tower stabilization beam **141**).

(21) FIG. 2 and FIG. 3 shows a detailed view of a tower section **150** being clamped to an interface plate **144** of a support flange **142** according to an exemplary embodiment of the present invention. The interface plate **144** is fixed by screw **301** to a foot and the support flange **142**, respectively. In a similar manner, the supporting system may also couple other support devices supporting the tower section **150**.

(22) The connection plate **100** has a platelike structure defining the tower connection portion **101**,

to which the tower section **150** can be mounted, and a support connection portion **102**, to which the support device **140** can be mounted. The tower connection portion **101** comprises a tower connection hole that can be matched or overlaid with the tower hole **151**, when the tower connection portion **101** is attached to a surface of the shell section **152** of the tower section **150**. Hence, the tower connection hole in the tower connection portion **101** extends along a radial direction  $r$  of the tower section **150**.

(23) The support connection portion **102** does also comprise a through hole, through which a further clamping fastener **211** (such as a screw or bolt) can be guided for fixing the support device **140** to the support connection portion **102**.

(24) The clamping device **210** comprises at least one clamping fastener which is guidable at least through the tower hole **151** and the tower connection hole.

(25) Specifically, the clamping device **110** is configured for detachably mounting the connection plate **100** to the tower section **150** and the support device **140**. Hence, a tower section **150**, which comprises radial directed tower holes **151** in the shell section **152**, can be detachably connected by the clamping device to a respective (temporary) support device **140**.

(26) In the shown example, the clamping device **110** comprises a plate washer **310** arrangeable at a surface of the shell section **152** of the turbine tower **150**. A clamping fastener **210** (for example a threaded bolt or a screw) can be screwed to the plate washer **310** and to the tower connection portion **101** of the connection plate **100** to thereby clamping the tower section **150** between the plate washer **310** and the tower connection portion **101**. The clamping fastener **210** is guided through the tower hole **151**, the tower connection hole and a hole in the plate washer **310**. The hole in the plate washer **310** may comprise a respective thread for providing a screw connection with the clamping fastener **210**.

(27) Additionally, an intermediate contacting plate **212** can be interposed between the shell section **152** and the connection portion **101** or a further intermediate contacting plate **311** can be interposed between the shell section **152** and the plate washer **310**. One surface of the contacting plates **212**, **311** may be adapted to the curvature of the shell section **152** of the tower section **150**. The contacting plates **212**, **311** may be made of a deformable and elastic material, such as bonded foam and/or hard rubber. The elastic material is compressed by the clamping device **110** and adopts the tower section curvature shape. The contacting plates **212**, **311** can be glued to the respective connection portions **101**, **102** of the connection plate **100**. The further contacting plate **311** may also be glued onto the plate washer **310**.

(28) The connection plate **100** has as an L-shape and the tower connection portion **101** and the support connection portion **102** are formed perpendicular to each other. Hence, a supporting hole or through opening of the support connection portion may have a hole extension perpendicular to the radial hole extension of the tower hole **151**.

(29) FIG. 4 shows a schematic view of an L-shaped connection plate **100** according to an exemplary embodiment of the present invention. The clamping device **110** comprises an adapter element **410** being adapted to the (oval) cross section of the tower hole **151** of the tower section **150**, wherein the adapter element **410** is insertable in the tower hole **151**. The adapter element **410** comprises a through hole extending between the plate washer **310** and the tower connection portion **101** of the connection plate **100** for inserting the clamping fastener **210**. The adapter element **410** is a solid body which fills the inner volume of the tower hole **151**. Hence, relative movements of the adapter element **410** inside the tower hole **151** and therefore also relative movements between the parts of the clamping device **110** and the connection plate **100** are reduced.

(30) The adapter element **410** comprises at least two clamping elements **411**, **412** for being clampable in the tower hole **151**. The clamping elements **411**, **412** are configured for being adjustable with respect to each other, such that the adapter element **410** is adaptable to the cross section of the tower hole. For example, the clamping elements **411**, **412** are movable with respect to each other. The clamping elements **411**, **412** comprise respective clamping surfaces which may be

pressed against the inner surfaces of the tower hole **151**.

(31) Specifically, the tower holes **151** in the shell section **152** are formed in a long hole, i.e., elongated hole, design. The adapter element **410** comprises a respective cross section for being for being insertable in a respective tower hole **151** having a respective long hole shaped cross section. Specifically, the above-described clamping elements **411**, **412** of the adapter element **410** are configured to be movable along a major axis of the elliptical or elongated hole cross section of the tower hole **151**.

(32) For example, in order to mount the support device, first the adapter elements **410** are inserted into the long hole shaped tower holes **151**, and the back plate washers **310** are fastened. For example, in a next step the tower support device **140**, such as the support flange **142** (UTTE) can then be mounted horizontally, like it would have been for an Lflange, and the remaining bolts, i.e., clamping fasteners **210**, can be fastened on the top side of the adapter element **410**. The tower section **150** can now be moved by lifting the support flange **142** with a tower mover.

(33) FIG. 5 shows a schematic view of an adapter element **410** according to an exemplary embodiment of the present invention. The adapter element **410** comprises a clamping rim **510** arranged at a top end and having a larger cross section than the main body part of the adapter element. Therefore, the clamping rim **510** has a larger cross section than the cross section of the tower hole **151**, such that the clamping rim **510** firms an attachment surface by which the clamping rim **510** can be laid onto a surface of the tower connection portion **101** and the shell section **152**, respectively. Hence, a further plate washer below the clamping fastener **210** is not necessary. The clamping rim **510** and the adapter element **410** may be monolithically formed of one piece. Alternatively, the clamping rim **510** and the adapter element **410** may be made of separate parts. Hence, the adapter element **410** and the clamping rim **510** may be e.g., welded together.

(34) FIG. 6 shows a schematic view of a supporting system according to an exemplary embodiment of the present invention. Specifically, FIG. 6 shows a back view of the embodiment shown in FIG. 1.

(35) A support flange **142** is shown which is coupled to a tower section handling device, in particular a transport vehicle or a support platform. The support flange **142** may have standardized connections with a transport device and thereby forms a Universal Tower Transport Equipment (UTTE). Furthermore, an interface plate **144** of the support flange **142** may provide a coupling area for the tower section **150** so that a plurality of connection plates **100** and clamping devices **110** of the supporting system can be used between the interface plate **144** and the tower section **150** to provide a stable support.

(36) Furthermore, a tower stabilization beam **141** is shown extending between two opposite sections of the shell section **152** of the tower section **150**. The tower stabilization beam **141** thereby enforces the tower section during transportation. The tower stabilization beam **141** functions as an Anti Ovalization Equipment (AO). For example, the connection plate and in particular the support connection portion **102** is mounted to an outer surface of the shell section **152** of the turbine tower **150**.

(37) Furthermore, a plurality of couplings to respective standard lashing elements **143** are shown. The lashing elements **143** are used to fix lashing strips for securing the tower section **150** e.g. from roll and pitch motions during sea and land transport.

(38) FIG. 7 shows a detailed view of the further plate washer **610** having an elongated hole **702** for fixing the interface plate **144** shown in FIG. 6.

(39) The support device **140**, in the example the interface plate **144** of the support flange **140** comprises a through opening **701**, wherein the clamping device **110** comprises a further plate washer **610** arrangeable at a surface of the interface plate **144**. The clamping device **110** further comprises a further clamping fastener **211** (for example a threaded bolt or a screw) for clamping the further plate washer **610** to the support connection portion **102** of the connection plate **100** and for thereby clamping the interface plate **144** between the further plate washer **701** and the support

connection portion **102**. The further clamping fastener **211** is guided through the coupling opening **701**, a supporting hole or through opening in the support connection portion **102** and a hole in the further plate washer **610**. The support connection portion **102** may comprise two through holes for fixing two further clamping fasteners **211**. The respective coupling opening **701** is large enough for guiding the two further clamping fasteners **211** through it.

(40) The hole in the further plate washer **610** comprises an elongated hole **702** through which the further clamping fastener **211** is guidable. Hence, the relative coupling positions may be adjusted such that a flexible fixation location can be provided.

(41) FIG. **8** shows a schematic view of a supporting system which couples a tower stabilization beam **141** to the tower section **150** according to an exemplary embodiment of the present invention. As shown in detail in FIG. **8**, a contacting plate **212** and the further plate washer **310** are arranged at the inner surface of the shell section **152** of the tower section **150**. In the shown example, in each second elongated tower hole **151** of the shell section **152**, a respective adapter element **410** of the clamping device **110** can be arranged. Hence, a robust removable connection between the tower section **150** and the tower stabilization beam **141** is provided.

(42) FIG. **9** shows a schematic view of a supporting system which couples lashing elements **143** to the tower section **150** according to an exemplary embodiment of the present invention. The lashing elements **143** are used to fix lashing strips to secure the tower section **150** from roll and pitch motions during sea and land transport.

(43) The support device **140** couples respective lashing elements **143** for fixing standard lashing brackets with attached lashing strips. The lashing element **143** is a thin plate, e.g. made of metal material, comprising two through holes, wherein in one through hole the further clamping fastener **211** is coupled and the second though hole is used for fixing a lashing bracket of the bracket strips. One of the two through holes of the support connection platform may be unused or may be used for coupling a second lashing element **143**.

(44) Although the present invention has been disclosed in the form of preferred embodiments and variations thereon, it will be understood that numerous additional modifications and variations could be made thereto without departing from the scope of the invention.

(45) For the sake of clarity, it is to be understood that the use of “a” or “an” throughout this application does not exclude a plurality, and “comprising” does not exclude other steps or elements.

## Claims

1. A supporting system for supporting a tower section of a wind turbine, the supporting system comprising a connection plate comprising a tower connection portion and a support connection portion, wherein the support connection portion is configured for being coupled to a tower support device for supporting the tower section having a cylindrical shape, wherein the tower connection portion comprises a tower connection hole for being coupleable to a tower hole extending radially outward along a radial direction through a circumferential flange provided on a cylindrical shell section of the tower section, a clamping device having at least one clamping fastener being partially insertable in and guidable through the tower connection hole of the tower connection portion and the tower hole for detachably clamping the connection plate to the tower section.
2. The supporting system according to claim 1, wherein the clamping device comprises a plate washer arrangeable at a surface of the shell section of the turbine, wherein the clamping device having the at least one clamping fastener for clamping the plate washer to the tower connection portion of the connection plate and for thereby clamping the tower section between the plate washer and the tower connection portion.
3. The supporting system according to claim 2, wherein the plate washer comprises a curved contact surface for contacting the tower section, wherein the curved contact surface is adaptable to the curvature of the respective surface of the tower section.

4. The supporting system according to claim 2, wherein the connection portion of the connection plate comprises a curved contact surface for contacting the tower section, wherein the curved contact surface is adaptable to the curvature of the respective surface of the tower section.
  5. The supporting system according to claim 2, wherein the clamping device comprises an adapter element being adapted to the cross section of the tower hole of the tower section, wherein the adapter element is insertable in the tower hole, wherein the adapter element comprises a through hole extending between the plate washer and the connection portion of the connection plate for inserting the at least one clamping fastener.
  6. The supporting system according to claim 5, wherein the adapter element comprises at least two clamping elements for being clampable in the tower hole, wherein the clamping elements are configured for being adjustable with respect to each other, such that the adapter element is adaptable to the cross section of the tower hole.
  7. The supporting system according to claim 5, wherein the adapter element comprises an elliptical cross section for being inserted in a respective tower hole having a respective elliptical cross section.
  8. The supporting system according to claim 1, wherein the connection plate has as an L-shape and the tower connection portion and the support connection portion are formed perpendicular to each other.
  9. The supporting system according to claim 1, wherein the tower support device comprises a coupling opening, wherein the clamping device comprises a further plate washer arrangeable at a surface of the support device, wherein the clamping device further comprises a further clamping fastener for clamping the further plate washer to the support connection portion of the connection plate and for thereby clamping the support device between the further plate washer and the support connection portion.
  10. The supporting system according to claim 9, wherein the further plate washer comprises an elongated hole through which the further clamping fastener is guidable.
  11. The supporting system according to claim 1, wherein the support device is a tower stabilization beam extending between two opposite sections of the shell section of the tower section.
  12. The supporting system according to claim 1, wherein the support device is a support flange being coupleable to a tower section handling device.
  13. The supporting system according to claim 12, wherein the tower section handling device is a transport vehicle or a support platform.
  14. A tower section supporting arrangement, comprising a tower section comprising a tower hole extending through a shell section of the tower section, a tower support device configured for supporting the tower section, and the system according to claim 1, wherein the support connection portion is coupled to the tower support device, wherein the clamping device is partially inserted in the tower connection hole of the tower connection portion and the tower hole for detachably clamping the connection plate to the tower section.
  15. A method of supporting a tower section of a wind turbine, the method comprising coupling a support connection portion of a connection plate to a tower support device, coupling a tower connection portion of the connection plate with a tower connection hole to a tower hole extending radially outward along a radial direction through a circumferential flange provided on a cylindrical shell section of the tower section having a cylindrical shape, and partially inserting and guiding a clamping device having at least one clamping fastener through and in the tower connection hole of the tower connection portion and the tower hole for detachably clamping the connection plate to the tower section.
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